

EDEXCEL INTERNATIONAL A LEVEL

WMA12/01 May/June R 2024 Worked Solutions

May/June R 2024 paper rebuilt with the worked solution after each question.

10

paper questions

WMA12

Pure 2

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - eliteigcse.com**P2**

PAST PAPER

WMA12/01 May/June R 2024

May/June R 2024 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Sequences & Series

1. The sequence $u_1, u_2, u_3, \dots$ satisfies

$$u_{n+2} = 3u_{n+1} - 2u_n$$

Given that

- $u_1 = 7$
- $u_3 = 4$

- (a) find the value of u_2

(2)

- (b) find $\sum_{r=1}^4 (u_r + 2r)$

(3)

Worked Solution - Question 1

Topic group

1. Use the recurrence at n equals one

$$u_3 = 3u_2 - 2u_1.$$

2. Substitute given values

$$4 = 3u_2 - 2(7), \text{ so } 4 = 3u_2 - 14.$$

3. Find u_2

$$3u_2 = 18, \text{ hence } u_2 = 6.$$

4. Find u_4

$$u_4 = 3u_3 - 2u_2 = 3(4) - 2(6) = 0.$$

5. Evaluate the sum

$$\sum_{r=1}^4 (u_r + 2r) = (7 + 2) + (6 + 4) + (4 + 6) + (0 + 8).$$

6. Add the terms

$$\text{The sum is } 9 + 10 + 10 + 8 = 37.$$

Final answer

$$(a) u_2 = 6.$$

$$(b) 37.$$

Question 2

Integration

2. The table shows corresponding values of x and y for a continuous curve with equation $y = f(x)$ between $x = -4$ and $x = 5$, where a is a constant.

x	-4	-2.5	-1	0.5	2	3.5	5
y	4.16	2.91	a	1.73	1.37	1.43	2.28

The trapezium rule is used with all the y values in the table to find an approximation for

$$\int_{-4}^5 f(x) \, dx$$

Given that the value of this approximation is 19.3

- (a) find the value of the constant a to 3 significant figures.

(3)

- (b) Use the given answer of 19.3 to find an approximate value for

$$\int_{-4}^5 (2f(x) - 3) \, dx$$

(2)

Worked Solution - Question 2

1. Use the trapezium width

The x values increase by 1.5, so $h = 1.5$.

2. Set up the trapezium estimate

$$19.3 = \frac{1.5}{2} [4.16 + 2.28 + 2(2.91 + a + 1.73 + 1.37 + 1.43)].$$

3. Solve for a

This simplifies to $19.3 = 15.99 + 1.5a$, so $a = \frac{3.31}{1.5} = 2.206\dots$

4. Round a

Therefore $a = 2.21$ to 3 significant figures.

5. Use the transformed integral

$$\int_{-4}^5 (2f(x) - 3)dx = 2 \int_{-4}^5 f(x)dx - \int_{-4}^5 3 dx.$$

6. Substitute the given estimate

This is $2(19.3) - 3(9) = 38.6 - 27 = 11.6$.

Final answer

(a) $a = 2.21$ to 3 s.f.

(b) 11.6.

Question 3

3. A circle has equation

$$x^2 + y^2 + 8x - 14y - 79 = 0$$

(a) Find

- (i) the coordinates of the centre of the circle,
- (ii) the radius of the circle.

(3)

Given that P is the point on the circle that is nearest the origin O ,

(b) find the exact length of OP

(2)

Worked Solution - Question 3

Topic group

1. Complete the square

$$x^2 + y^2 + 8x - 14y - 79 = 0 \text{ becomes}$$
$$(x + 4)^2 - 16 + (y - 7)^2 - 49 - 79 = 0.$$

2. Write standard form

$$(x + 4)^2 + (y - 7)^2 = 144.$$

3. Read centre and radius

The centre is $(-4, 7)$ and the radius is **12**.

4. Find distance from centre to origin

The distance from $(-4, 7)$ to $O(0, 0)$ is $\sqrt{(-4)^2 + 7^2} = \sqrt{65}$.

5. Use origin inside the circle

Since $\sqrt{65} < 12$, the origin is inside the circle.

6. Find nearest distance to the circle

The nearest point on the circle lies along the line from the centre to the origin, so $OP = 12 - \sqrt{65}$.

Final answer

(a)(i) $(-4, 7)$.

(a)(ii) **12**.

(b) $12 - \sqrt{65}$.

Question 4

Binomial Expansion

4. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(3 + 2x)^6$$

giving each coefficient in simplest form.

(4)

- (b) Hence find the coefficient of x^2 in the expansion of

$$\left(2x^2 - \frac{1}{6x}\right)(3 + 2x)^6$$

(3)

Worked Solution - Question 4

1. Write the expansion terms

$$(3 + 2x)^6 = 3^6 + \binom{6}{1}3^5(2x) + \binom{6}{2}3^4(2x)^2 + \binom{6}{3}3^3(2x)^3 + \dots$$

2. Simplify the first four terms

This gives $729 + 2916x + 4860x^2 + 4320x^3$.

3. Identify x squared contributions

In $\left(2x^2 - \frac{1}{6x}\right)(3 + 2x)^6$, the x^2 term comes from $2x^2$ times the constant term and $-\frac{1}{6x}$ times the x^3 term.

4. Calculate the coefficient

The coefficient is $2(729) - \frac{1}{6}(4320) = 1458 - 720 = 738$.

Final answer

(a) $729 + 2916x + 4860x^2 + 4320x^3$.

(b) 738.

Question 5

Trigonometric Equations

5.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

The depth of water, D metres, in a harbour on a particular day is given by the equation

$$D = 8 + 5 \sin\left(\frac{\pi t}{6} + 3\right) \quad 0 \leq t < 24$$

where t is the number of hours after **midnight**.

- (a) Show that the depth of water in the harbour at 2 am is just over 4 metres. (1)
- (b) Find, to the nearest minute, the first time after **midday** when the depth of water in the harbour is exactly 6 metres. (5)

Worked Solution - Question 5

1. Substitute t equals two

$$D = 8 + 5 \sin \left(\frac{\pi(2)}{6} + 3 \right) = 8 + 5 \sin \left(\frac{\pi}{3} + 3 \right).$$

2. Evaluate

$D = 4.064 \dots$, so the depth at 2 am is just over 4 metres.

3. Set depth equal to six

$$6 = 8 + 5 \sin \left(\frac{\pi t}{6} + 3 \right), \text{ so } \sin \left(\frac{\pi t}{6} + 3 \right) = -\frac{2}{5}.$$

4. Use the time interval

After midday means $t > 12$. Then $\frac{\pi t}{6} + 3$ is greater than $2\pi + 3$.

5. Choose the first suitable angle

The first angle after $2\pi + 3$ with sine $-\frac{2}{5}$ is $3\pi + \sin^{-1} \left(\frac{2}{5} \right)$.

6. Solve for t

$$\frac{\pi t}{6} + 3 = 3\pi + \sin^{-1} \left(\frac{2}{5} \right), \text{ so } t = 13.056 \dots \text{ hours after midnight.}$$

7. Convert to clock time

$0.056 \dots$ hours is about 3.38 minutes, so the time to the nearest minute is **13:03**, or **1:03 pm**.

Final answer

(a) $D(2) = 4.06 \dots$ m, just over 4 m.

(b) 13:03, or 1:03 pm.

Question 6

Applications of Differentiation

6.

$$f(x) = 4x^3 + px^2 + 8x + q$$

where p and q are constants.

Given that

- $(2x + 3)$ is a factor of $f(x)$
- $f(x)$ has a remainder of -5 when divided by $(x + 2)$

(a) (i) show that $p = 10$

(ii) find the value of q .

(5)

(b) Hence find the range of values of x for which $f(x)$ is decreasing.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

Worked Solution - Question 6

Topic group

1. Use the factor theorem

Since $(2x + 3)$ is a factor, $x = -\frac{3}{2}$ is a root.

2. Substitute x equals negative three halves

$$4\left(-\frac{3}{2}\right)^3 + p\left(-\frac{3}{2}\right)^2 + 8\left(-\frac{3}{2}\right) + q = 0.$$

3. Simplify the first condition

$$-\frac{27}{2} + \frac{9}{4}p - 12 + q = 0, \text{ so } \frac{9}{4}p + q = \frac{51}{2}.$$

4. Use the remainder theorem

The remainder on division by $(x + 2)$ is $f(-2)$.

5. Substitute x equals negative two

$$f(-2) = 4(-8) + 4p - 16 + q = 4p + q - 48.$$

6. Use the given remainder

$$4p + q - 48 = -5, \text{ so } 4p + q = 43.$$

7. Solve for p and q

Subtract $\frac{9}{4}p + q = \frac{51}{2}$ from $4p + q = 43$: $\frac{7}{4}p = \frac{35}{2}$, so $p = 10$. Then $4(10) + q = 43$, so $q = 3$.

8. Differentiate

With $p = 10$, $f(x) = 4x^3 + 10x^2 + 8x + 3$, so $f'(x) = 12x^2 + 20x + 8$.

9. Factor the derivative

$$f'(x) = 4(3x + 2)(x + 1).$$

10. Find where decreasing

The quadratic derivative is negative between its roots, so $f(x)$ is decreasing for $-1 < x < -\frac{2}{3}$.

Final answer

(a)(i) $p = 10$.

(a)(ii) $q = 3$.

(b) $-1 < x < -\frac{2}{3}$.

Question 7

Trigonometric Equations

7.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

- (i) Solve, for $0 \leq x < 2\pi$, the equation

$$3 \sin x \tan x = 11 + \cos x$$

giving the answers in radians to 3 decimal places.

(5)

- (ii) Given that

- $0 < \theta < 90^\circ$
- $\cos \theta = \frac{1}{3}$

find, in simplest form, the exact value of $\tan \theta$

(2)

Worked Solution - Question 7

1. Rewrite $\tan x$

$$3 \sin x \tan x = 11 + \cos x \text{ becomes } \frac{3 \sin^2 x}{\cos x} = 11 + \cos x.$$

2. Multiply by $\cos x$

$$3 \sin^2 x = 11 \cos x + \cos^2 x.$$

3. Use sine squared

$$3(1 - \cos^2 x) = 11 \cos x + \cos^2 x, \text{ so } 4 \cos^2 x + 11 \cos x - 3 = 0.$$

4. Solve for cosine

$$(4 \cos x - 1)(\cos x + 3) = 0, \text{ so } \cos x = \frac{1}{4} \text{ or } \cos x = -3.$$

5. Reject impossible value

$$\cos x = -3 \text{ is impossible, so } \cos x = \frac{1}{4}.$$

6. Find x values

For $0 \leq x < 2\pi$, this gives $x = 1.318 \dots$ or $x = 4.965 \dots$, so $x = 1.318, 4.965$ radians.

7. Use the right triangle identity

$$\text{Given } 0 < \theta < 90^\circ \text{ and } \cos \theta = \frac{1}{3}, \sin \theta = \sqrt{1 - \frac{1}{9}} = \frac{2\sqrt{2}}{3}.$$

8. Find $\tan \theta$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{2\sqrt{2}/3}{1/3} = 2\sqrt{2}.$$

Final answer

(i) $x = 1.318, 4.965$.

(ii) $2\sqrt{2}$.

Question 8

Arithmetic Sequences

8. (i) (a) In an **arithmetic** series the first term is a and the common difference is d .

Show that

$$S_n = \frac{n}{2} \{2a + (n-1)d\} \quad (3)$$

- (b) Hence find

$$900 + 892 + 884 + \dots + 500 \quad (3)$$

- (ii) Given that the first three terms of a **geometric** series are

$$k+4 \quad k-2 \quad 11-k$$

where k is a constant,

- (a) show that

$$2k^2 - 11k - 40 = 0 \quad (3)$$

Given also that this series is convergent,

- (b) find the value of S_∞ (4)

Worked Solution - Question 8

1. Write the first and last terms

The first term is a and the n th term is $a + (n - 1)d$.

2. Use the arithmetic sum formula

$$S_n = \frac{n}{2}(\text{first term} + \text{last term}).$$

3. Show the formula

$$\text{Therefore } S_n = \frac{n}{2}\{a + [a + (n - 1)d]\} = \frac{n}{2}\{2a + (n - 1)d\}.$$

4. Identify the arithmetic series

For $900 + 892 + 884 + \cdots + 500$, the first term is 900 , the common difference is -8 , and the last term is 500 .

5. Find the number of terms

$$500 = 900 + (n - 1)(-8), \text{ so } n - 1 = 50 \text{ and } n = 51.$$

6. Find the sum

$$S = \frac{51}{2}(900 + 500) = 35700.$$

7. Use the geometric condition

For $k + 4$, $k - 2$, $11 - k$ to be consecutive geometric terms,
 $(k - 2)^2 = (k + 4)(11 - k).$

8. Show the quadratic

$$k^2 - 4k + 4 = -k^2 + 7k + 44, \text{ so } 2k^2 - 11k - 40 = 0.$$

9. Solve for k

$2k^2 - 11k - 40 = 0$ gives $k = 8$ or $k = -\frac{5}{2}$.

10. Use convergence

If $k = 8$, the terms are **12, 6, 3** and the common ratio is $\frac{1}{2}$, so the series is convergent. If $k = -\frac{5}{2}$, the common ratio is -3 , so it is not convergent.

11. Find S infinity

Use $k = 8$: $S_{\infty} = \frac{12}{1 - 1/2} = 24$.

Final answer

(i)(a) Proven.

(i)(b) 35700.

(ii)(a) Proven.

(ii)(b) $S_{\infty} = 24$.

Question 9

Laws of Logarithms

9. Given that

$$3 \log_2(t+4) - 2 \log_2(t-2) = 7$$

(a) verify that $t = 4$ is a solution of the above equation,

(2)

(b) show that

$$t^3 - 116t^2 + 560t - 448 = 0$$

(3)

(c) Hence, using algebra and showing your working, solve

$$3 \log_2(t+4) - 2 \log_2(t-2) = 7$$

giving each answer in simplest form.

(Solutions based entirely on calculator technology are not acceptable.)

(4)

Worked Solution - Question 9

Topic group

1. Verify t equals four

Substitute $t = 4$: $3 \log_2(8) - 2 \log_2(2) = 3(3) - 2(1) = 7$.

2. Combine the logarithms

$$3 \log_2(t + 4) - 2 \log_2(t - 2) = \log_2 \left(\frac{(t + 4)^3}{(t - 2)^2} \right).$$

3. Use seven as a logarithm

$$7 = \log_2(128), \text{ so } \frac{(t + 4)^3}{(t - 2)^2} = 128.$$

4. Clear the denominator

$$(t + 4)^3 = 128(t - 2)^2.$$

5. Expand

$$t^3 + 12t^2 + 48t + 64 = 128t^2 - 512t + 512.$$

6. Show the cubic

$$\text{Rearranging gives } t^3 - 116t^2 + 560t - 448 = 0.$$

7. Use the known root

Since $t = 4$ is a root, $(t - 4)$ is a factor.

8. Factor the cubic

$$t^3 - 116t^2 + 560t - 448 = (t - 4)(t^2 - 112t + 112).$$

9. Solve the quadratic

$$t^2 - 112t + 112 = 0 \text{ gives } t = \frac{112 \pm \sqrt{112^2 - 4(112)}}{2} = 56 \pm 12\sqrt{21}.$$

10. Use the log domain

The original equation requires $t > 2$. The value $56 - 12\sqrt{21}$ is less than 2 , so it is rejected. Hence $t = 4$ or $t = 56 + 12\sqrt{21}$.

Final answer

(a) Verified.

(b) Proven.

(c) $t = 4$ or $t = 56 + 12\sqrt{21}$.

Question 10

Applications of Differentiation

10.

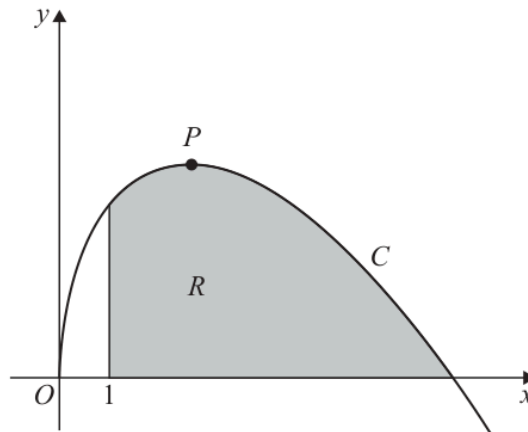


Figure 1

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

Figure 1 shows a sketch of part of the curve C with equation

$$y = \frac{9x - x^2}{2\sqrt{x}} \quad x > 0$$

The point P is the stationary point on C .

(a) Find, using calculus, the x coordinate of P .

(4)

The finite region R , shown shaded in Figure 1, is bounded by the curve C , the x -axis and the line with equation $x = 1$

(b) Using calculus, calculate the exact area of R .

(5)

Worked Solution - Question 10

Topic group

1. Rewrite the curve

$$y = \frac{9x - x^2}{2\sqrt{x}} = \frac{1}{2}(9x^{1/2} - x^{3/2}).$$

2. Differentiate

$$\frac{dy}{dx} = \frac{1}{2} \left(\frac{9}{2}x^{-1/2} - \frac{3}{2}x^{1/2} \right) = \frac{3(3-x)}{4\sqrt{x}}.$$

3. Use the stationary condition

For $x > 0$, the denominator is not zero. So $\frac{dy}{dx} = 0$ when $3 - x = 0$.

4. Find P

Therefore the x coordinate of P is 3.

5. Find where the curve meets the x-axis

$y = 0$ when $9x - x^2 = 0$, so $x = 9$ for the positive intercept.

6. Set up the area

The region is under the curve from $x = 1$ to $x = 9$, so $R = \int_1^9 \frac{9x - x^2}{2\sqrt{x}} dx$.

7. Integrate

$$R = \int_1^9 \frac{1}{2}(9x^{1/2} - x^{3/2})dx = \left[3x^{3/2} - \frac{1}{5}x^{5/2} \right]_1^9.$$

8. Evaluate

$$R = \left(3(27) - \frac{1}{5}(243) \right) - \left(3 - \frac{1}{5} \right) = \frac{148}{5}.$$

Final answer

(a) $x = 3$.

(b) $\frac{148}{5}$.